Enrollment No: 2105022022

MID-TERM EXAMINATION
(Course Name -B.Tech ECE/ECE-AI) (Semester- 3rd)
(October, 2023) OFF LINE mode

Subject Code: BEC 205 | **Subject: Network Analysis And Synthesis**

Time : 1 ½Hours | **Maximum Marks :** 30

Note: Q. 1 is compulsory.

Q1			
	(a). Using source transformation technique, covert the voltage source shown in fig-1 into equivalent current source.	**(2.5*4)**	
	(b). Write down the initial conditions of R, L and C at t=0 and t=infinity		
	(c). Write down the condition for maxima in DC circuit ad AC circuit		
	(d). Write down the expression for Thevenin's Voltage and Thevenin's Resistance given by Millman's Theorem		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a). For the ckt shown I fig-2, determine the current I through the 10 ohm resistance using Norton's Theorem		
	(b). For the ckt shown I fig-2, determine the current I through the 10 ohm resistance using Mesh Analysis Method		
	(c). For the ckt shown I fig-2, determine the current I through the 10 ohm resistance using Supper- Position Theorem		
Q3	**(Attempt any one Parts) UNIT-2**	**(10)**	
	(a) Consider the RLC series ckt shown in fig-3, Vs=2V: R=6 ohms : L=2 H ; C=0.25 F. Determine i(0) ; di(0)/dt ; $d^2i(0)/dt^2$; i(t)		
	(b). Consider a series RLC ckt with the capacitor initially charged to a voltage of 1V as indicated in the fig-4. Find the expression for i(t). Also draw the S-domain representation of the ckt.		

For Figure, please turn over.

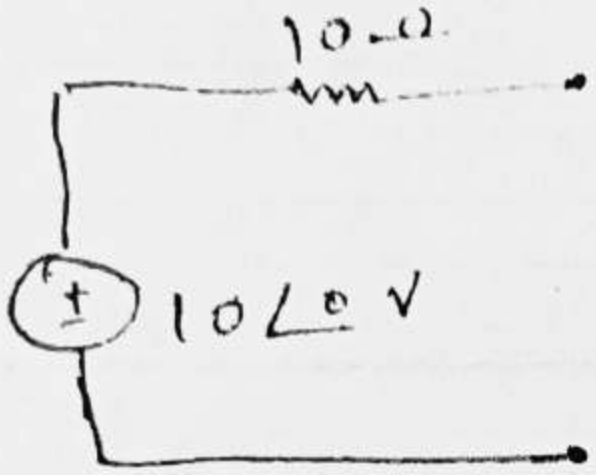


fig-1.

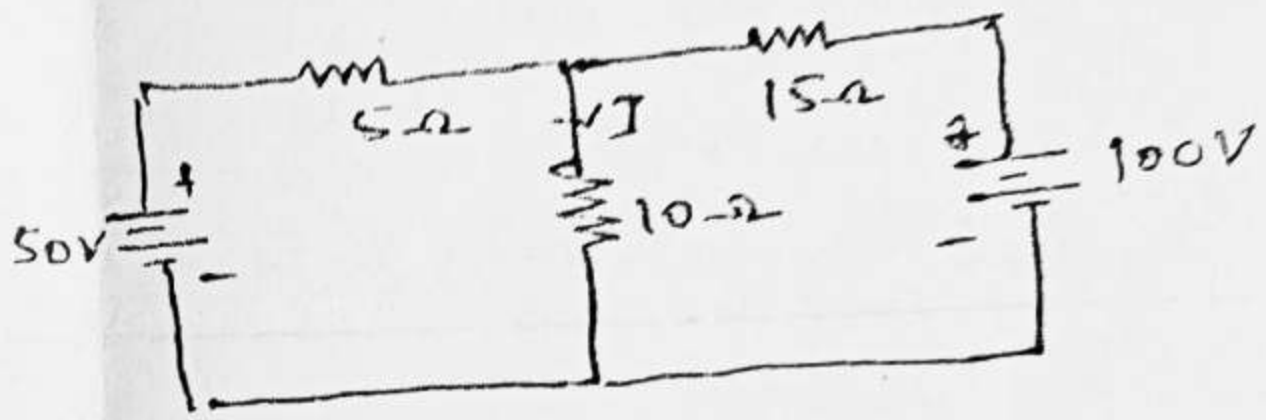


fig-2

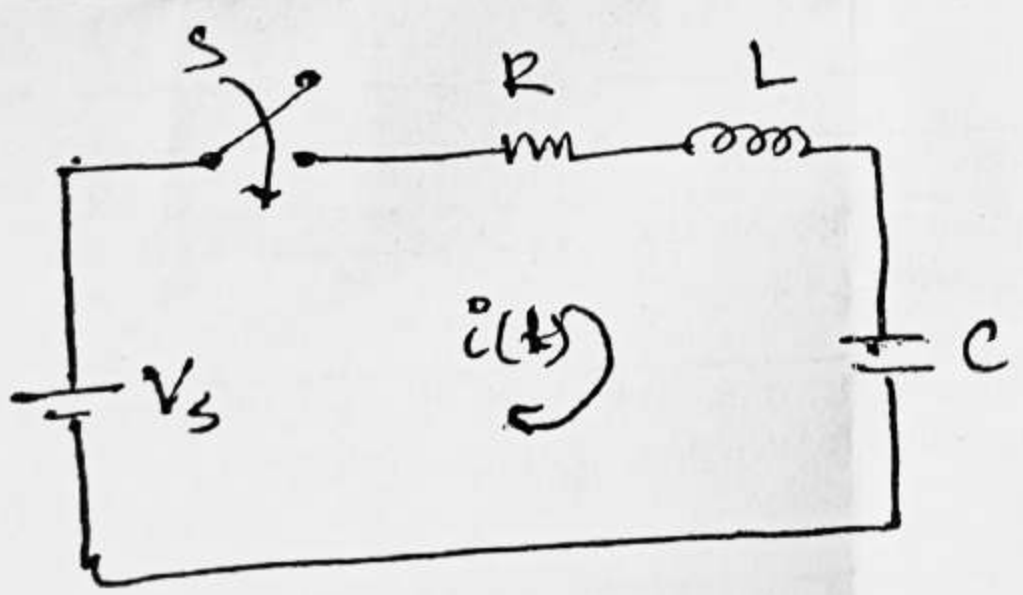


fig-3

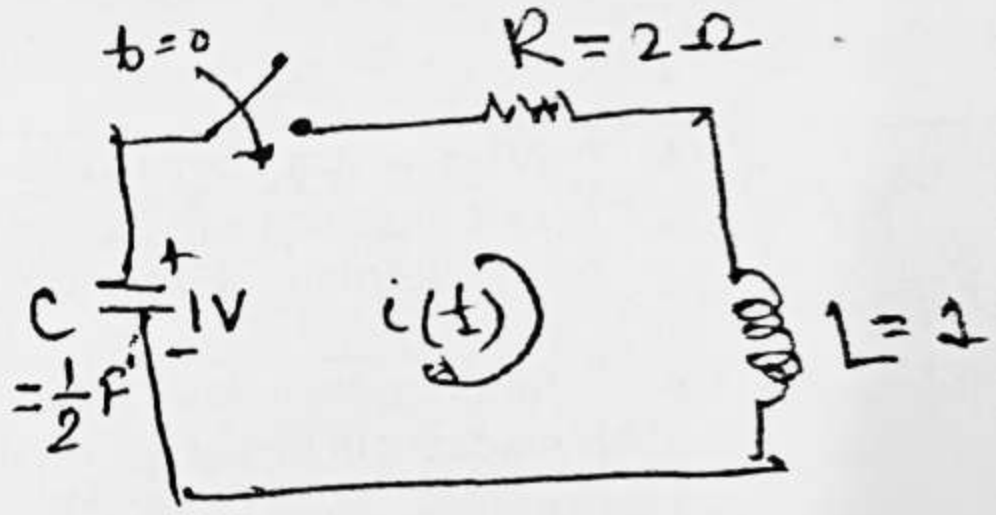


fig-4